AI-Powered Project Tracking Report

Automatically generated visual summary of project tracking datasets.

AI Summary

1. Milestone delivery is slipping materially, especially on the critical path
- Description: Milestones are completing later than planned on average, with a meaningful tail of large delays.
- Evidence: In `milestones.csv`, `slip_days` has a mean of **+2.58 days** across **341 completed/recorded milestones**, with a maximum slip of **42 days**. In `tasks.csv`, task `slip_days` also shows delay pressure overall: mean **-0.47 days**, but the distribution is wide with a maximum slip of **22 days** and 75th percentile at **+2 days**. Several task labels include **“Critical Path”**, indicating some delay risk is directly affecting schedule-critical work.
- Impact: Even a small average slip can accumulate across dependent milestones and push project end dates, especially where critical-path tasks are involved. A few extreme slips can absorb schedule float and create cascading delays.
- Recommended Action: Review all milestones with slip greater than **5 days**, prioritize those tied to critical-path tasks, and re-baseline only after confirming recovery actions. Add weekly recovery tracking for the worst-slipping projects and require owners to publish mitigation dates, not just status updates.

2. Overdue and blocked tasks are a major execution risk
- Description: The task dataset shows a meaningful population of work that is either overdue or blocked, which suggests active execution bottlenecks.
- Evidence: `tasks.csv` includes an `is_overdue` flag for all **1,800 tasks**, and status values include **Blocked** and **Not Started** in addition to **In Progress** and **Completed**. The sample rows already show overdue in-progress tasks such as `T00002` (`is_overdue = True`, `status = In Progress`, `slip_days` not yet recorded) and `T00005` (`is_overdue = True`, `status = In Progress`, `remaining_effort_hours = 25`). The `risk_score` distribution is also elevated overall: mean **0.313**, with the top quartile above **0.428** and maximum **1.0**.
- Impact: Overdue tasks with remaining effort still open are the clearest sign of schedule slippage turning into delivery risk. If blocked tasks are not cleared quickly, they will continue to inflate downstream delays and resource churn.
- Recommended Action: Create a daily exception review for all tasks where `is_overdue = True` or `status = Blocked`, starting with high-priority items and tasks with high `risk_score`. Require each blocked task to have a named blocker owner, next action, and target unblock date.

3. Resource load is above the balanced range for many people, creating a likely bottleneck
- Description: Resource allocation data shows average monthly load above the balanced band, with many people likely overcommitted.
- Evidence: In `resource_conflicts.csv`, `total_allocated_hours` averages **184.6 hours/month**, above the stated balanced band of **150–170 hours/month**. The 75th percentile is **220.75 hours**, and the maximum is **260 hours**. The dataset also includes severity categories such as **Critical**, **High**, **Medium**, and **Underallocated**, indicating the platform is already flagging capacity imbalance.
- Impact: Sustained over-allocation usually leads to slower task completion, more context switching, and higher defect/rework risk. It also makes schedule recovery harder because the same constrained people are likely on the critical path.
- Recommended Action: Rebalance work from people above **170 hours/month** to underallocated staff before adding overtime. Focus first on people supporting overdue or high-risk tasks, and use the `severity` field to prioritize the most constrained resources.

4. Issue and risk exposure is concentrated in schedule-impacting supplier and capacity problems
- Description: The issue log shows a recurring pattern of supplier delay and capacity constraint problems with non-trivial schedule impact.
- Evidence: In `issues_and_risks.csv`, `estimated_schedule_impact_days` averages **4.54 days**, with a maximum of **21 days**. The median is **4 days**, and the 75th percentile is **7 days**, so the impact is not limited to a few outliers. The sample includes a **Major** `Supplier Delay` issue with **12 estimated schedule impact days** and

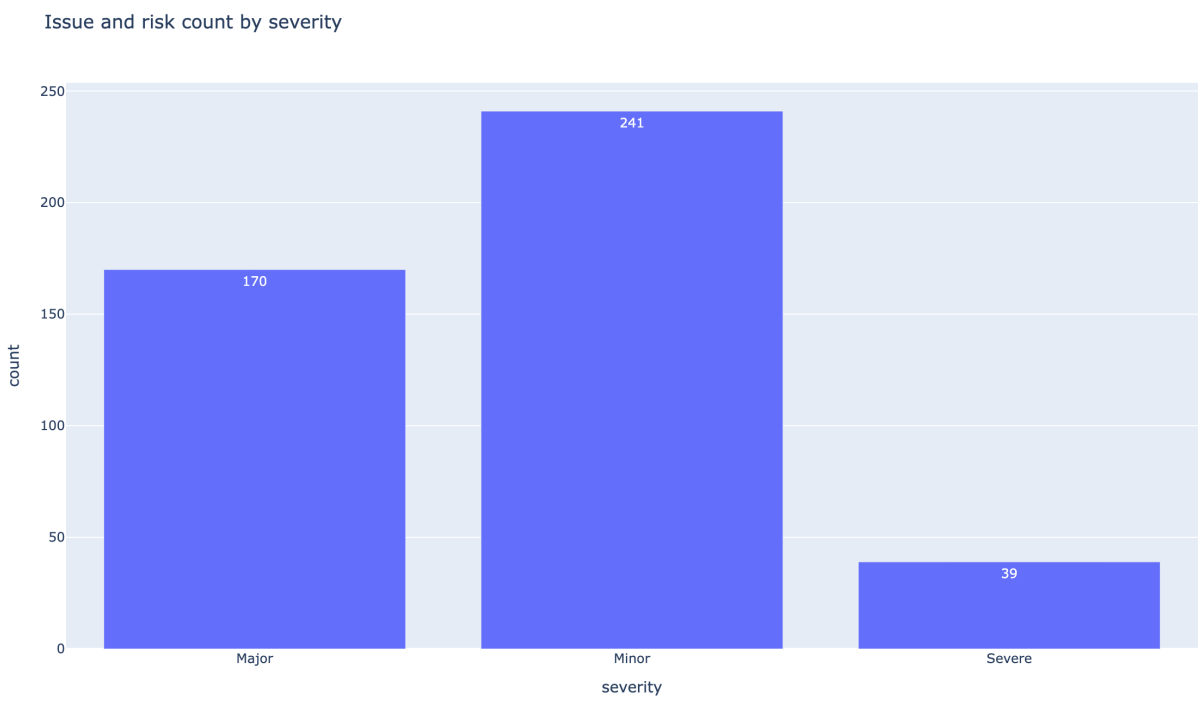
status **Investigating** (`ISS00005`), and other entries show `Capacity Constraint` affecting workstreams such as Design and Site Installation.
- Impact: Supplier and capacity issues are especially dangerous because they often affect multiple downstream tasks and milestones, not just one work item. If they remain open, they can drive both schedule slips and resource conflicts.
- Recommended Action: Escalate all open or investigating issues with **estimated_schedule_impact_days >= 7** into a weekly risk review. For supplier-related risks, confirm alternate sourcing or expediting plans; for capacity constraints, assign backup resources and adjust sequencing immediately.

5. Notifications suggest the project is generating too many escalation-worthy events, but follow-through may be weak
- Description: The notification stream contains multiple high-urgency triggers, yet acknowledgements are not visible in the sample rows, which may indicate weak response discipline.
- Evidence: `notifications.csv` has **900 rows** and **8 trigger types**, including **Milestone Slip**, **Resource Conflict**, **High Risk Score**, **Task Overdue**, **Blocked Task**, **Scope Change**, **Issue Opened**, and **Issue Escalated**. Urgency levels include **Action Required** and **Escalation**. In the first five records shown, all `acknowledged` values are **False**, including an **Action Required** high-risk-score notification and an **Escalation** milestone-slip notification.
- Impact: If escalation notifications are not acknowledged promptly, the project may be signaling risk without getting the response needed to correct it. That weakens governance and allows issues to persist longer than necessary.
- Recommended Action: Track acknowledgement rate by trigger type and urgency, and require response SLAs for **Action Required** and **Escalation** notifications. If acknowledgement is consistently low, simplify the alerting process and assign explicit owners for each notification category.

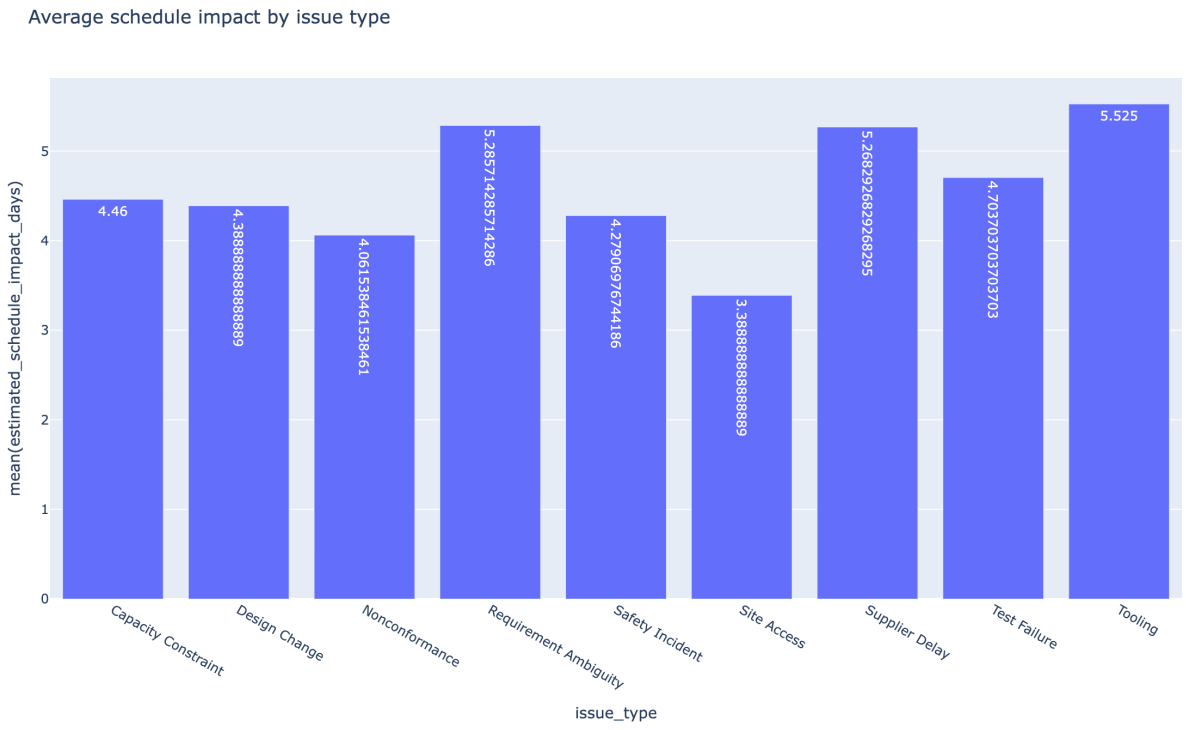
6. Schedule and risk management should be tightened at the project level, not just task level
- Description: The project portfolio shows a mix of budgets, complexity, and baseline dates, but the execution signals suggest some projects need closer governance than others.
- Evidence: `projects.csv` contains **30 projects** with `complexity` values including **High** and **Medium**, budgets ranging from **4.21 to 26.41 MUSD** in the sample rows, and steering committee ownership listed for each project. Since tasks, milestones, issues, and notifications all reference `project_id`, the data supports project-level rollups, and the presence of repeated high-risk triggers implies some projects are accumulating multiple execution problems.
- Impact: Without project-level prioritization, managers may react to individual issues while missing the few projects that are driving most of the schedule and resource pressure.
- Recommended Action: Rank projects by a combined view of overdue tasks, milestone slip, open issues, and resource conflicts. Put the top-risk projects on a weekly steering committee agenda and require a single recovery plan per project with clear owners and dates.

Dataset: Project_Data/issues_and_risks.csv

Issue and risk count by severity

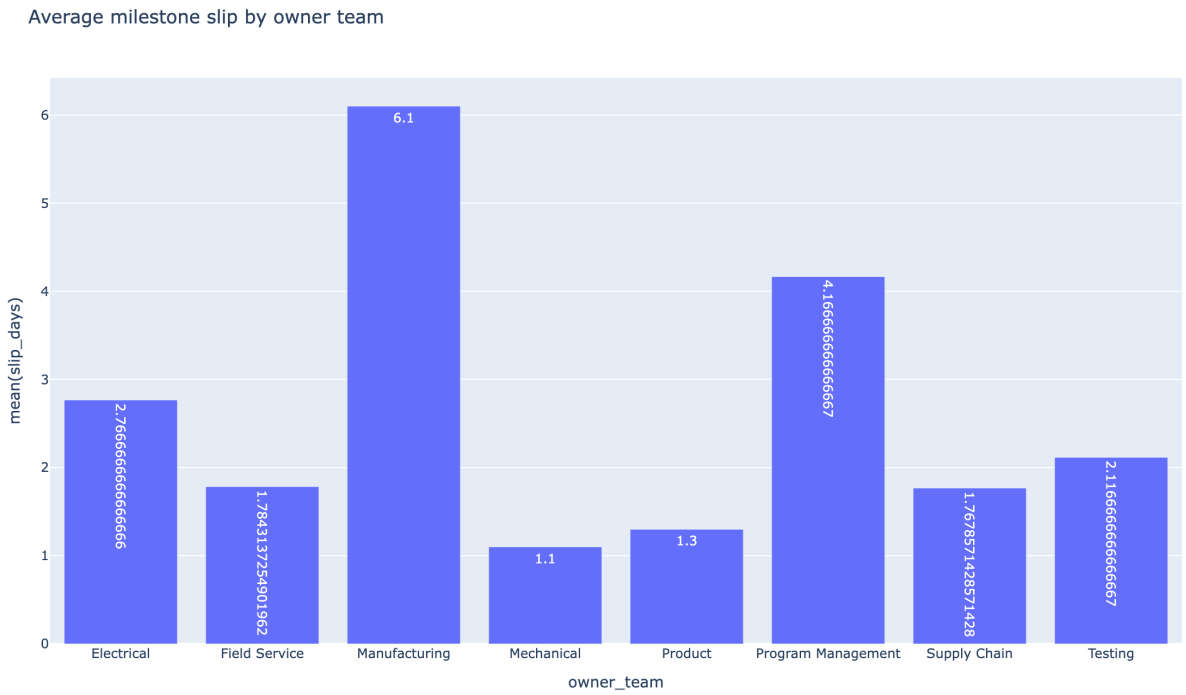


Average schedule impact by issue type



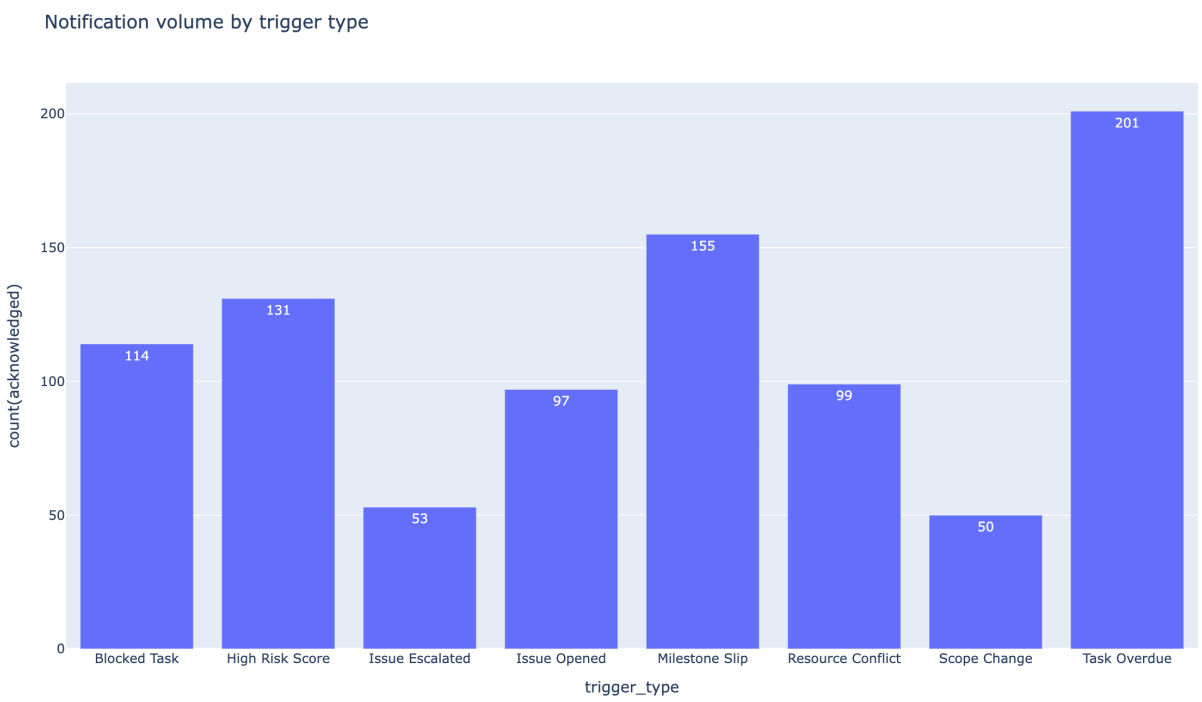
Dataset: Project_Data/milestones.csv

Average milestone slip by owner team

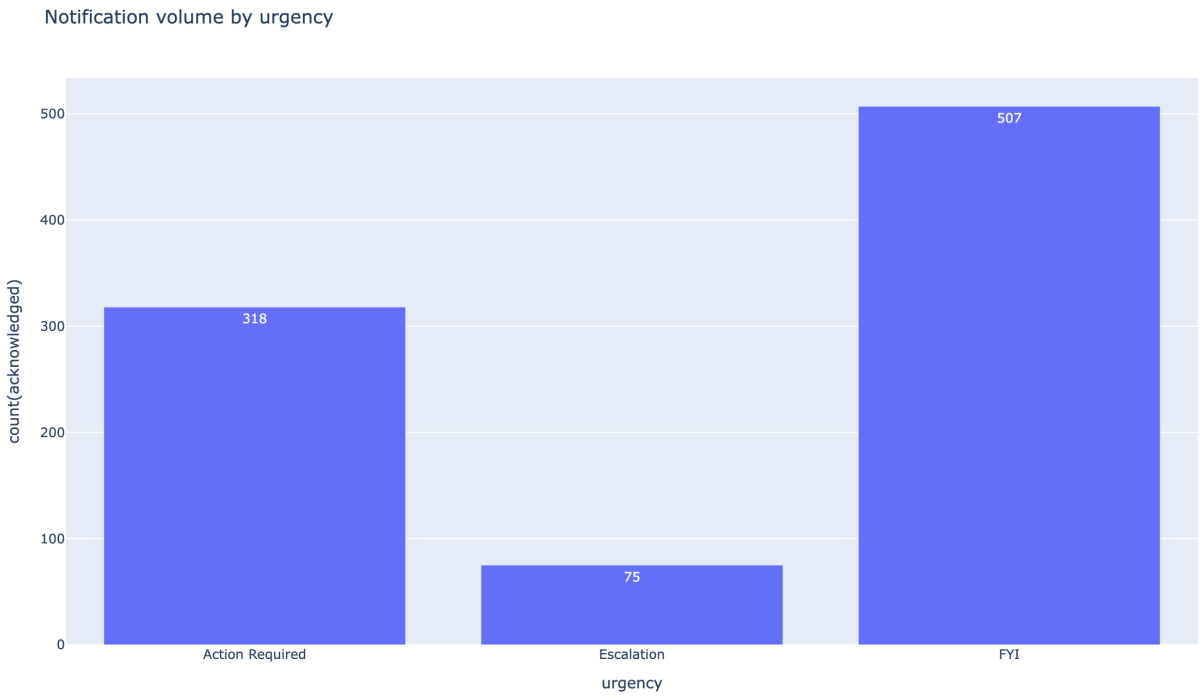


Dataset: Project_Data/notifications.csv

Notification volume by trigger type

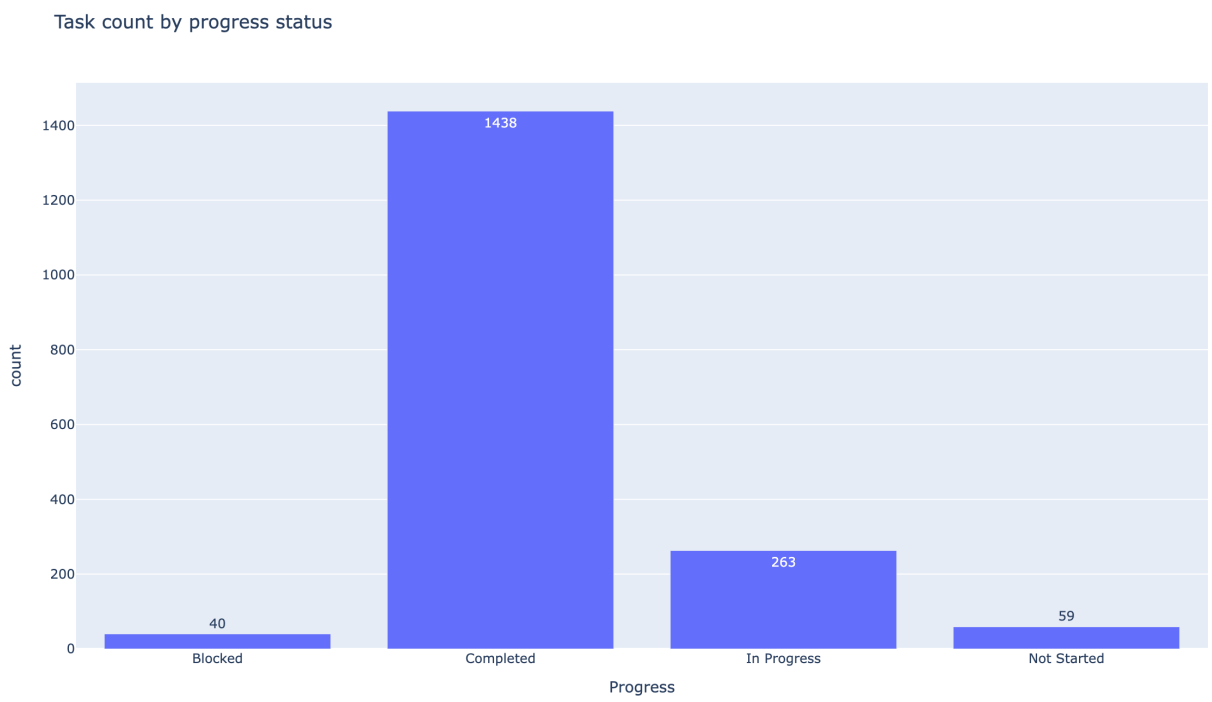


Notification volume by urgency

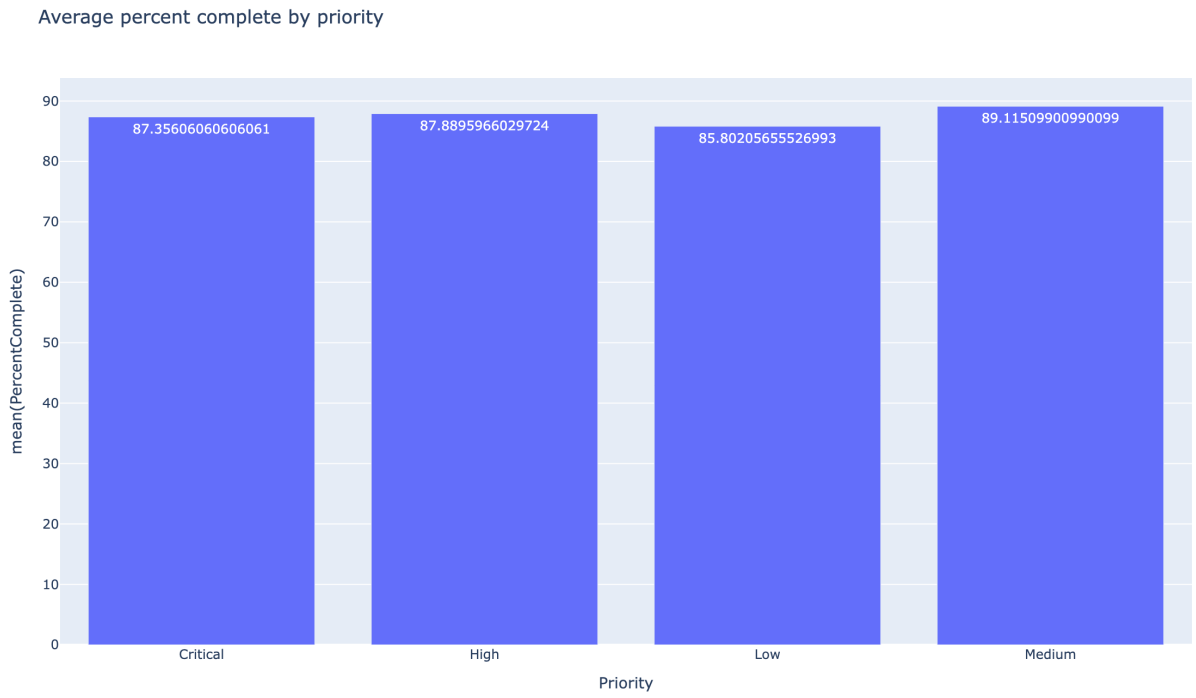


Dataset: Project_Data/planner_export_like.csv

Task count by progress status

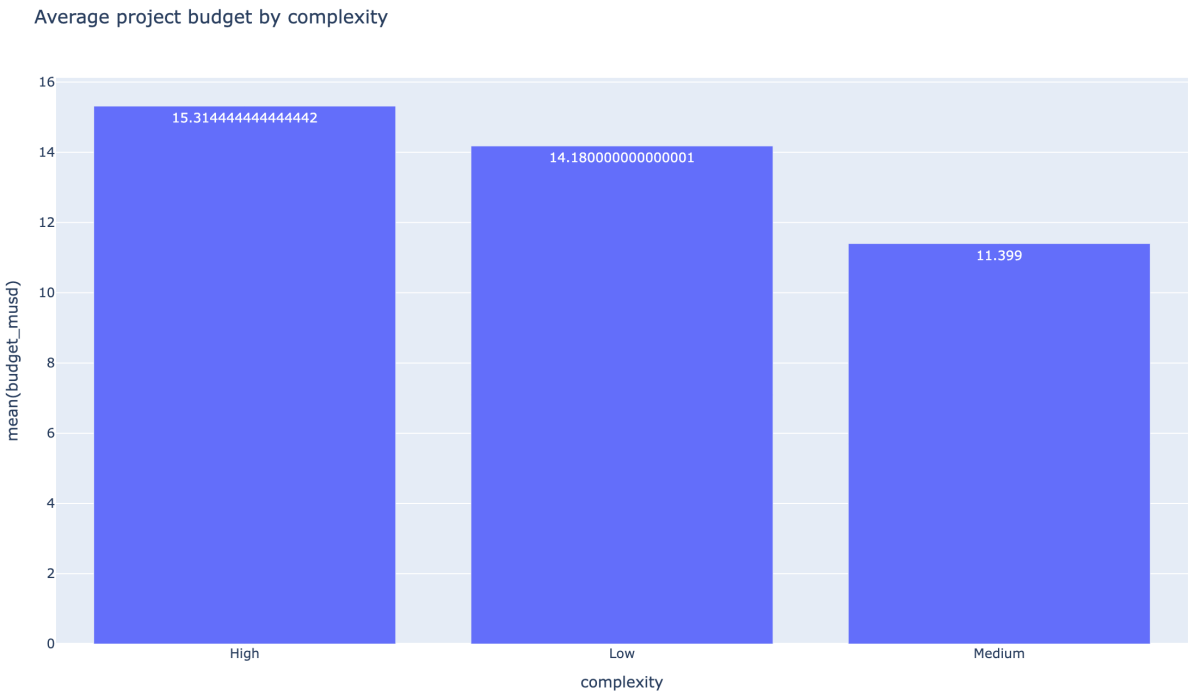


Average percent complete by priority



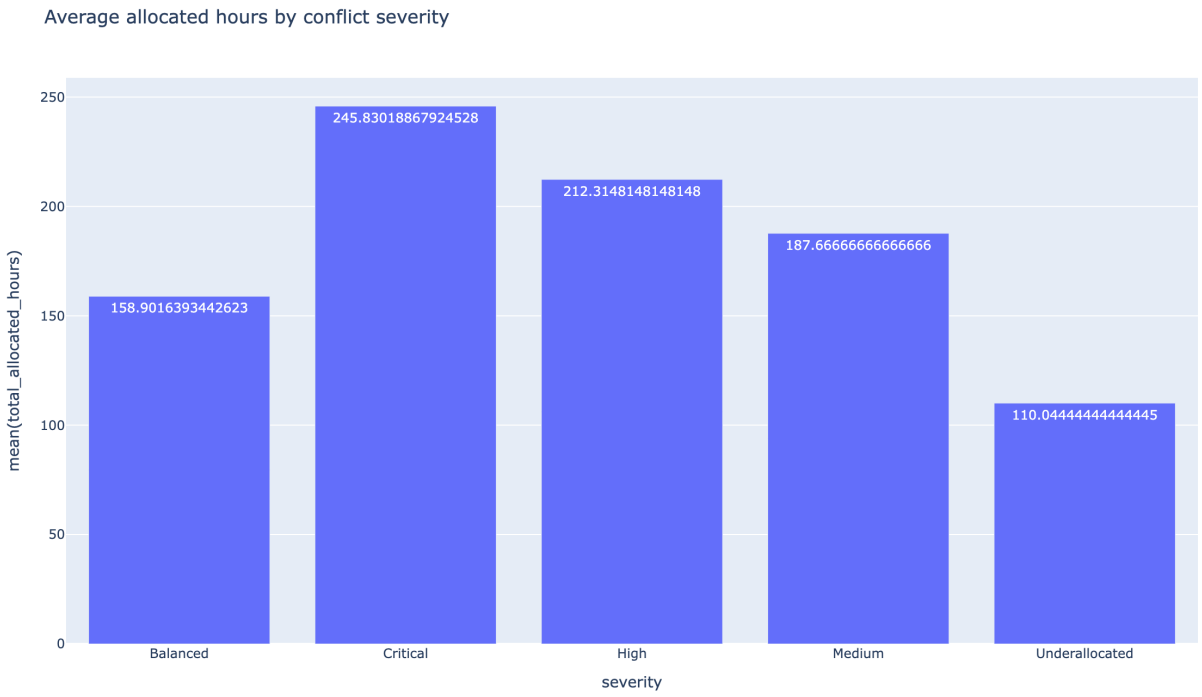
Dataset: Project_Data/projects.csv

Average project budget by complexity

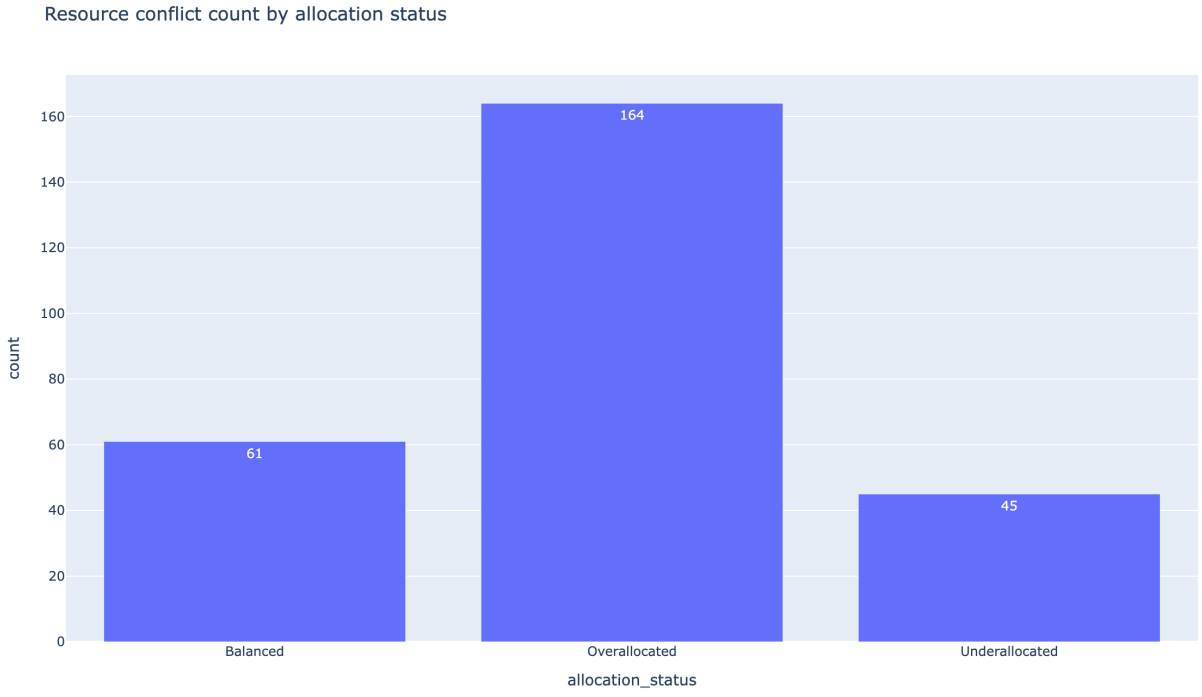


Dataset: Project_Data/resource_conflicts.csv

Average allocated hours by conflict severity

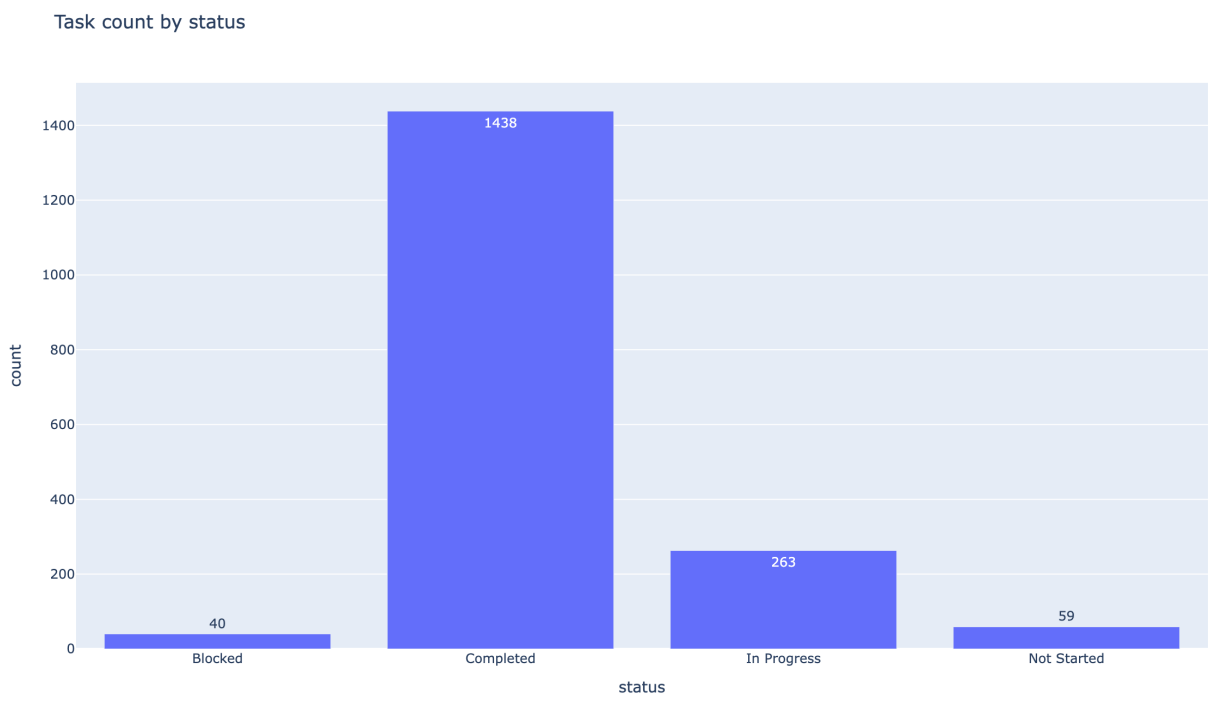


Resource conflict count by allocation status



Dataset: Project_Data/tasks.csv

Task count by status



Overdue task count by priority

